

Propagation Mechanics of an Embedded Agent: How Testimony, Divergence, and Inquiry Arise Without Being Represented

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Abstract

We study what an agent embedded in a finite world *does* when it is engaged, and we show that a family of behaviours an observer would call *testimony*, *divergent report* (including what is ordinarily named a lie), *rumour*, and *questioning* arise from the agent’s ordinary activity *without the agent representing any of them*. We take an agent to have a purpose it pursues and a positive cost on every act, and to access its world only by *search*: to recall or report anything is to run a bounded walk over a finite graph of what it has individuated. From four axioms—a positive floor on the cost of any distinction, purpose-directed search, a positive floor on the cost of any act, and the non-completeness of the world—we prove, as graph and order theorems, that: **(T1)** recalling and searching are one operation, so every report an agent gives is freshly searched, never fetched; **(T2)** two searches from the same start to the same conclusion traverse different interiors and neither can reconstruct the other’s route (*path opacity*), so an agent cannot certify its own account against any standard; **(T3)** the same event registers a different result in agents with different internal graphs, and each registration is a correct individuation in its own graph, so divergence of report between agents is not error in either; **(T4)** no agent can read the value another’s report would have to match, so the property an observer names “the account is false” is undefined *inside* any agent and is only ever a relation an outside party computes between two accounts—the acting agent holds no

representation of it; **(T5)** an account passed from agent to agent acquires each relay’s search as a composed factor, so its content drifts multiplicatively along a chain with no relay altering it on purpose; and **(T6)** because no report resolves to a held value, an exchange cannot close, so a further act—which an observer names a question—is forced by the residual gap and is selected by that gap, not by any represented intent to inquire. We then locate the vocabulary: we prove that “lie,” “rumour,” and “question” are predicates on *relations between accounts*, computable by an observer or a second agent, and that none of them names any state or act of the agent that produced the account. The agent contains no concept of deceiving, gossiping, or asking; the concepts live entirely on the side of whoever watches. We give constructions, an observer-side audit, and a numerical study on explicit agents. The reading of the formal objects as testimony and inquiry is marked as such; no theorem depends on it.

Keywords: embedded agent; search; recognition–search identity; path opacity; receiver-relative report; unextractable value; catalytic drift; forced response; testimony; emergence without representation; non-player character.

1 Introduction

1.1 The question

An agent placed in a world and engaged by another party must produce something: a recollection, a report, an answer. We ask what it

is *doing* when it does this, and we arrive at an observation that reorganises the usual picture. Behaviours that an onlooker readily names—the agent *testified*, it *lied*, it *spread a rumour*, it *asked a question*—turn out not to be things the agent does deliberately, nor things it needs any concept of in order to do. They are names an onlooker attaches, after the fact, to the output of the agent’s ordinary activity. The agent that “lied” did nothing structurally different from the agent that “told the truth.” The vocabulary is on the onlooker’s side. Testimony, lying, and communicative meaning have a large philosophical literature [3, 12, 10, 20, 6]; our contribution is to show these predicates need no representation *inside* the agent.

This paper makes that observation precise and proves it. We show that an embedded agent, equipped only with a purpose and a cost on its acts, and accessing its world only by bounded search, *emits* testimony, divergent report, rumour, and questioning *without representing any of them*—and that the very properties by which an onlooker classifies these outputs are undefined inside the agent and are only ever relations computed from outside.

1.2 The claim: emergence without representation

Our central claim is one of *emergence*, deliberately not one of necessity. We do not prove that such an agent must, in any given world, produce a lie or a question; that would require assumptions about what the world supplies. We prove the cleaner and more defensible thing: *these outcomes occur without the agent representing them*. Nowhere in the agent’s machinery is there a concept of deceiving, of gossiping, or of asking; yet its outputs are correctly so classified by an observer. We then show *where* the classifying vocabulary can attach at all, and find—as a result, not a premise—that it attaches only to relations between accounts, never to any state or act of the agent that produced one.

1.3 Method

Every object below is a finite weighted graph or a construction on one, and every result is a theorem from four stated axioms and standard mathematics [1, 7]. We define an agent’s

world as a graph of what it has individuated, its report as a walk on that graph, and its purpose and act-cost as in a companion development which we restate here so the paper stands alone. The deductions are combinatorial (cuts and walks), order-theoretic (monotone histories), and use only elementary facts about minimum cuts, which are computable exactly by maximum flow [9, 13, 8, 4]. Where a tempting claim could not be proved at this standard it has been removed; every interpretive reading is confined to a remark on which no theorem depends.

1.4 Organisation

Section 2 fixes the agent, its world-graph, its purpose and act-cost, and states the axioms. Section 3 proves that recalling is searching (T1). Section 4 proves path opacity and that an agent cannot certify its own account (T2). Section 5 proves receiver-relative report and that divergence is not error (T3). Section 6 proves that falsity is undefined inside the agent and locates it as an observer-side relation (T4). Section 7 proves catalytic drift of a relayed account (T5). Section 8 proves that a further act is forced by the residual gap (T6). Section 9 gathers the result: where each name lives. Section 10 reports a numerical study. Section 11 discusses scope.

2 The agent and its world

We build the agent’s world as a graph of what it has told apart, and its report as a search over that graph.

Definition 2.1 (World-graph; medium). The *world-graph* of an agent is a finite weighted graph $\Gamma = (V, E, w)$ with $w : E \rightarrow \mathbb{R}_{>0}$, together with a distinguished vertex $\mathbf{m} \in V$, the *medium*, adjacent to every other vertex. The vertices $v \neq \mathbf{m}$ are the *items* the agent has individuated (the things it has told apart); an edge is a *contact*, its weight the cost of the separation distinguishing its endpoints. The graph is connected. The medium is the single vertex standing for “everything else,” against which each item is told apart—individuation against a background being the classical problem of reference and identity [15, 14].

Remark 2.2 (The world-graph is a finite window on an infinite whole). The world-graph is not constructed by the agent or handed to it as input; it is what the agent has *individuated so far* of a whole that exceeds it. The agent is a finite subgraph of an infinite whole, and the world-graph Γ is the portion of that whole visible from the agent’s position—the items it has told apart, together with the contacts among them. The medium m is not a separate object bolted on for bookkeeping: it *is* the rest of the infinite whole, always already present, the undifferentiated “everything else” against which each item is told apart. On this reading non-compleatability (Axiom 2.10) is not an extra hypothesis about the world growing but a direct consequence of the whole being genuinely infinite: no finite stage of individuation exhausts it, so there is always a further item, and the medium never empties. This grounds the world-graph in a foundational account of the agent as a proper part of an infinite whole, and removes any appearance that Γ is an arbitrary given.

Definition 2.3 (Cut; separation cost; resting cut). For $\emptyset \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$ with weight $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$. The *separation cost* of an item v is the minimum weight of a cut placing v with the medium excluded,

$$\sigma(v) = \min_{S \ni v, m \notin S} w(\partial(S)),$$

attained on finite V ; a minimising side is a *resting cut* of v and is the v – m minimum cut, computable by maximum flow via the max-flow/min-cut theorem [9, 13].

We now restate the agent’s purpose and act-cost, so the paper is self-contained.

Definition 2.4 (Disposition space; purpose). The agent has a compact convex *disposition space* $\mathcal{B} \subseteq \mathbb{R}^n$ and a continuously differentiable, λ -strongly convex *drive* $\Phi : \mathcal{B} \rightarrow \mathbb{R}$ ($\lambda > 0$) [2]. Its *purpose* is the unique minimiser $x^* = \arg \min_{x \in \mathcal{B}} \Phi(x)$, the attractor of the descent $\dot{x} = -\nabla \Phi(x)$ [11]; the *residual to purpose* at x is $D(x) = \Phi(x) - \Phi(x^*) \geq 0$.

Definition 2.5 (Act; act-cost). An *act* of the agent (a step of search, a committed report) draws from a finite repertoire, each act carrying a *cost* $c > 0$. A bounded agent com-

mits only finitely many acts per unit time, in the sense of bounded rationality and resource-limited agency [18, 19, 16].

The four axioms follow. They are meant to hold of any bounded agent embedded in a world it did not finish building.

Axiom 2.6 (Distinction floor). No separation is free: there is $\beta > 0$ with $w(e) \geq \beta$ for every contact. Hence $\sigma(v) \geq \beta > 0$ for every item (a v – m cut is nonempty by connectedness).

Remark 2.7 (The distinction floor is not an independent axiom). We state Axiom 2.6 for convenience, but it is *not* independent of the others: it follows both from what a distinction *is* (Theorem 2.11 below, via boundary thickness) and, quantitatively, from finiteness at each stage. We keep it as a named axiom so later sections may cite the floor directly, but the reader should regard it as a derived fact, demoted to the status of a lemma by Theorem 2.11. The genuinely primitive commitments are purpose-directed search (Axiom 2.8), the act floor (Axiom 2.9), and non-compleatability (Axiom 2.10).

Axiom 2.8 (Purpose-directed search). The agent accesses its world only by search directed at its purpose: to recall, report, or answer is to run a walk on Γ whose steps are selected to reduce the residual D toward x^* (Definition 3.1 below). The agent has no direct read-out of any item’s value independent of such a walk.

Axiom 2.9 (Act floor). No act is free: there is a floor on act-cost, $c \geq \beta_{\text{act}} > 0$, so the agent commits only finitely many acts per unit time.

Axiom 2.10 (Non-completable world). The world is non-completable: for every finite stage of the agent’s world-graph there is a further item not yet individuated. This is an order condition (no last stage), not a claim about cardinality.

Theorem 2.11 (The floor is constitutive of distinction, and attained). *Every separation of the agent’s world-graph carries strictly positive weight, and the floor $\beta > 0$ is attained:*

- (i) **Positive by boundary thickness.** *A separation of zero cost is a boundary of zero thickness, and a zero-thickness boundary is not a boundary—it fails to separate. Hence every contact that actually tells its*

endpoints apart has positive weight, and each item's resting cut, being a genuine separation from the medium, has $\sigma(v) > 0$.

- (ii) **Attained by finiteness at each stage.** *At any finite stage the agent's world-graph has finitely many contacts, so $\beta := \min_e w(e)$ is a minimum over a finite set of positive reals, hence attained and strictly positive; and $\sigma(v) \geq \beta > 0$ for every item since a v -m cut is nonempty by connectedness.*

Thus Axiom 2.6 is not an independent stipulation but a consequence of what a distinction is (i) together with finiteness (ii).

Proof. (i) To hold two items apart at zero cost is to hold them apart by a boundary across which nothing is expended to tell one side from the other; but a boundary that costs nothing to cross does not divide its two sides—they are not, at zero cost, told apart, hence not two. So any contact that genuinely separates has positive weight. An item's resting cut is by Definition 2.3 the minimum v -m cut, a genuine separation of v from “everything else,” so it too has positive total weight: $\sigma(v) > 0$. (ii) The world-graph at any finite stage has a finite edge set (each edge an individuated contact); the infimum of finitely many positive reals is a minimum, attained and strictly positive, so $\beta := \min_e w(e) > 0$ is well-defined at that stage and, being a minimum, is not a loose bound. Connectedness makes every v -m cut nonempty, so $\sigma(v) \geq \beta > 0$. The two together give Axiom 2.6 as a derived fact. $\square$

Remark 2.12 (What non-completeness does, and does not, establish). It is worth separating two roles the reader might conflate. The *floor* $\beta > 0$ is established by Theorem 2.11—by what distinction is, not by non-completeness. What Axiom 2.10 does is different and equally load-bearing: it guarantees the world-graph is always *expandable*—there is always a further item to individuate, so the agent's graph never closes into a finite catalogue. This is why no individuation resolves to a residue-free point (Section 6): not because each individuation “costs something” (that is the floor's job) but because the comparison against the whole never terminates, there being always more world. The floor keeps each

separation positive; non-completeness keeps the world open. Both are needed, and they are different facts.

Remark 2.13 (Why the floor matters here). The floor is the single fact the whole paper turns on. Because every individuation costs at least β , no item is ever resolved to a zero-content point (Section 6); because every act costs at least β_{act} , search is bounded and its route is a committed, non-repeatable history (Section 4). Everything an observer later names—testimony, divergence, drift, inquiry—is a consequence of the gap the floor keeps open.

3 Recalling is searching: no report is fetched

We first show that, for an agent of this kind, there is no distinction between recalling something and searching for it: to produce a report is always to run a fresh search, never to fetch a stored value.

Definition 3.1 (Query; search process; report). A *query* to the agent is a target item x^* (what the engagement is about). A *search process* for x^* from a seed item v_0 is a D -descending walk $W = (v_0, v_1, \dots, v_k)$ on Γ ending at a resting cut that individuates the agent's answer. The *report* (equivalently the *account*) $\mathfrak{a}(W)$ is the terminal resting cut together with the item it individuates: what the agent returns.

Theorem 3.2 (Recognition–search identity). *For an agent accessing its world only by search (Axiom 2.8), recalling an item and searching for it are one operation: producing the report for a query x^* is running a search process for x^* , and there is no distinct “fetch” that returns a stored value without a walk.*

Proof. By Axiom 2.8 the agent has no read-out of an item's value independent of a D -descending walk; the only operation that yields a report is such a walk (Definition 3.1). A “fetch” would be a value returned with no walk, i.e. a zero-act access; by Axiom 2.9 every act costs $\geq \beta_{\text{act}} > 0$, so a zero-act access returns nothing. Hence every report is the terminus of a search, and recalling is searching. $\square$

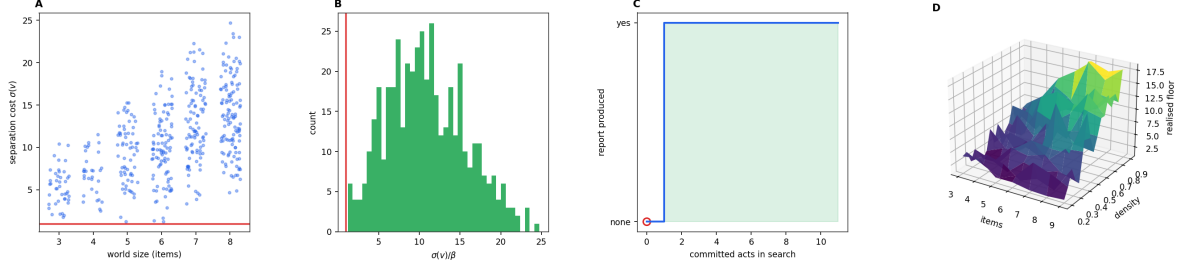


Figure 1: The Floor and Recall-Is-Search: No Item Is Individuated for Free, and No Report Is Fetched. (A) Separation cost $\sigma(v)$ on the linear y -axis versus world size in items on the linear x -axis (3 to 9), every item of 80 random worlds (steel-blue), with the floor line marked (crimson). Every separation cost, computed as an exact minimum cut, sits on or above the floor: the empirical content of Theorem 2.11, that identifying an item carries an irreducible residual $\sigma(v) \geq \beta > 0$. (B) Histogram of the normalised separation cost $\sigma(v)/\beta$ on the linear x -axis with the unit line marked (crimson), counts on the linear y -axis (sea-green). The support begins at 1 and the smallest observed value is 1.092—the floor is close to tight, so the residual the whole paper turns on is real and small. (C) Report produced (none / yes) on the y -axis versus committed acts in the search on the linear x -axis (0 to 12) (steel-blue step, positive region shaded sea-green), with the zero-act case ringed (crimson). A report appears only after at least one committed act, the empirical content of Theorem 3.2: recalling is searching and a zero-act fetch returns nothing. (D) Three-dimensional realised-floor surface over world size in items (x -axis, 3 to 9) and graph density (y -axis, 0.2 to 0.9) (viridis colourmap), viewing angle elevation 24° azimuth -58° . The realised floor stays strictly positive across the grid: no configuration individuates any item for free.

That proof, resting on Axiom 2.8, shows only that search-access is what this agent has. The substantive question is *why* fetch-access is not merely absent but incoherent for an agent of this kind—why it could not be added. The next theorem answers it, and does not depend on Axiom 2.8.

Theorem 3.3 (Fetch-access is architecturally incoherent under a monotone history). *Suppose, contrary to Axiom 2.8, the agent could fetch: return a queried item’s value as it was stored at some earlier stage, with no walk on the current graph. Then the returned value is incoherent with the agent that returns it. For the world-graph changes with every committed act—each act costs $\geq \beta_{\text{act}} > 0$ (Axiom 2.9) and increments the monotone committed record, which never decreases (Corollary 4.5), so at the moment of the fetch the graph Γ is strictly later than the graph at which the value was stored. A stored value is an individuation against the earlier graph and its medium; returned now, it is an answer appropriate to a prior agent—one the current agent cannot re-occupy, since the record does not decrease (monotone non-return, Corol-*

lary 4.5). Fetch therefore delivers an account the present agent could not itself have produced and cannot own; it is architecturally incoherent for an agent with a monotone history, not merely excluded by fiat.

Proof. Each committed act increments the monotone record (Axiom 2.9, Corollary 4.5) and alters the state in which any subsequent report is produced; so the graph and history at the fetch strictly succeed those at storage. A fetched value $\mathbf{a}_0$ was the terminus of an individuation against the stored-stage graph Γ_0 and its medium $\mathbf{m}_0$. To be the current agent’s own account it would have to be an individuation against the current Γ and $\mathbf{m}$; but $\Gamma \neq \Gamma_0$ (the record has grown) and by Corollary 4.5 the agent cannot return to the stage of Γ_0 . Hence $\mathbf{a}_0$ is the account of a prior, now-unreachable agent, not of the one holding it: fetch returns a value the present agent can neither have generated nor re-occupy the standpoint of. The incoherence is structural—it follows from monotone non-return alone—so fetch cannot be coherently added to an agent with a growing record,

independently of Axiom 2.8. $\square$

Corollary 3.4 (Every report is freshly produced). *No two engagements return a pre-stored answer; each runs its own search and produces its report anew. In particular the agent’s committed history grows by at least one act per report (Axiom 2.9), so the state in which a report is produced is never exactly a prior state.*

Proof. By Theorem 3.2 each report is a search terminus, and by Axiom 2.9 each search commits ≥ 1 act of positive cost, incrementing the agent’s committed act-count. Two engagements therefore occur at distinct act-counts, so their producing states differ. $\square$

Remark 3.5 (Interpretation: no canned lines). Theorem 3.2 is the formal content of the intuition that a convincing embedded agent does not read from a script. What it says when engaged is searched, not fetched; and because the search runs against the agent’s current world-graph and purpose, the same query at two moments need not return the same account. We use only the graph statement.

4 Path opacity: an agent cannot certify its own account

We now show that two searches reaching the same conclusion take different routes, and that neither the endpoints nor any agent can recover which route was taken. The consequence is that an agent has no access to a “how I got here” that it could check against a standard—it cannot certify its own account.

Definition 4.1 (Committed record; interior of a search). The *committed record* of a search $W = (v_0, \dots, v_k)$ is $M(W) = k$, the number of steps committed. The *interior* of W is the sequence of intermediate items $(v_1, \dots, v_{k-1})$ it passed through, excluding the seed and terminus.

Definition 4.2 (Agent state: terminus and count, not the walk). The agent’s *state* is the pair (v_k, M) of its current item (the terminus of its last search) and its committed record M . The state carries the *terminus* and the *count*; it does not carry the interior $(v_1, \dots, v_{k-1})$ of the walk that produced the terminus. An agent

retains where it arrived and how many acts it has committed, not the route it took to arrive.

Remark 4.3 (Why the state excludes the interior). This is a substantive premise, made explicit: were the agent to store its full walk history, it could in principle recover its route, and the no-self-certification corollary below would fail. The premise is warranted by the act floor: retaining each of the k interior items would itself be k further committed acts of positive cost, growing without bound as searches accumulate; an agent with a finite act-budget per unit time (Axiom 2.9) cannot carry the interior of every past search. The economical and forced state is the terminus-plus-count of Definition 4.2, and the results of this section rest on it.

Theorem 4.4 (Path opacity). *Let W_1, W_2 be two search processes with the same seed v_0 and the same terminal report $\mathbf{a}$. Then:*

- (i) *their interiors may differ, and in a connected world-graph with any item of degree ≥ 2 generically do; and*
- (ii) *the report $\mathbf{a}$ (seed and terminus) does not determine the interior: from $\mathbf{a}$ alone one cannot recover which of W_1, W_2 was run.*

Proof. (i) If some item on a v_0 -terminus route has degree ≥ 2 , there are two admissible continuations from it (two contacts), yielding two walks with the same endpoints and different interiors; connectedness supplies at least one such branching item whenever the graph is not a bare path. (ii) The report $\mathbf{a}$ records only the seed and the terminal resting cut; the interiors of W_1, W_2 differ (i) while their $\mathbf{a}$ coincide, so the map interior $\mapsto \mathbf{a}$ is not injective and has no inverse: $\mathbf{a}$ does not determine the interior. $\square$

Corollary 4.5 (An agent cannot reconstruct its own route). *The committed record grows monotonically (each step adds one and none is removed, since undoing a step is itself a further step), so once a search terminates, its interior is a past state the agent does not re-occupy. The agent holds its report $\mathbf{a}$, not the walk that produced it; by Theorem 4.4(ii) $\mathbf{a}$ does not encode the interior. Hence the agent cannot reconstruct the route by which it reached its own account.*

Proof. Monotonicity of the record: a step increments it; an “undo” is a further step incre-

menting it again, so it never decreases and the interior is not re-entered. By Definition 4.2 the agent’s state is the terminus-plus-count (v_k, M) and does *not* include the interior; Theorem 4.4(ii) shows the retained report $\mathbf{a}$ underdetermines the interior. With the interior neither carried in the state (Definition 4.2, Remark 4.3) nor recoverable from $\mathbf{a}$, the route is unavailable to the agent. $\square$

Corollary 4.6 (No self-certification). *An agent cannot certify its own account against any standard that would require its route: it holds the report but not the search that produced it, and cannot recover the search. Whatever an observer might mean by “the agent knows how it reached this,” the agent does not.*

Proof. Certifying a route requires possessing it; by Corollary 4.5 the agent does not possess and cannot recover its route. Hence no route-dependent certification is available to the agent. $\square$

Remark 4.7 (Why this is the crux of “definition-free”). Corollary 4.6 is the technical heart of the paper’s thesis. A concept of lying would require the agent to compare its account against how things stand and note a divergence it is responsible for. But the agent cannot even recover its own route (Corollary 4.5), let alone measure it against a standard it does not possess (Section 6). The machinery an agent would need in order to *represent* its account as false is simply absent. This is not a design omission; it is forced by path opacity and the monotone record.

5 Divergent report is not error in either agent

Two agents engaged about the same event return different accounts. We show this is not a fault in either: each account is a correct individuation in its producer’s own world-graph, and there is no shared value of which the two are competing approximations.

Definition 5.1 (Registration). The *registration* of an external event in an agent is the resting cut the event lands in within that agent’s

world-graph: its individuation against *that* agent’s medium (its own “everything else”).

Theorem 5.2 (Receiver-relative report). *The same external event registers different resting cuts in agents with different world-graphs, and each registration is a correct individuation at its own agent’s floor. No registration is privileged as the account; there is no shared point-value of which the registrations are approximations.*

Proof. The registration is the event’s resting cut in the agent’s graph (Definition 5.1); different graphs give different cuts (a sparse graph individuates the event against few contrasts, a dense graph against many), so the registrations differ. Each is the minimum v – m cut in its own graph, hence a correct floor-attaining individuation there. Were there a privileged shared value, a registration differing from it would be incorrect; but each registration is the correct individuation in its own graph, and—since completing an individuation to a residue-free point would require exhausting the non-completable world (Axiom 2.10, Remark 2.12)—no such shared point-value exists. Hence none is privileged. $\square$

Corollary 5.3 (Divergence is not error). *Two agents’ accounts of one event may differ while each is a correct individuation in its own world-graph. The divergence is not a mistake in either agent; it is the difference of their world-graphs, and neither account is the account.*

Proof. Immediate from Theorem 5.2: each registration is correct in its own graph, and there is no privileged value relative to which either could be wrong. $\square$

Remark 5.4 (Two distinct claims, and which is earned here). It is important to separate two claims that are easy to run together. *First:* divergence between two honest agents’ accounts is not error in either. This is Corollary 5.3, and it is fully earned by Theorem 5.2 (T3)—each account is correct in its own graph, with no privileged value relative to which either is wrong. *Second:* a true account and a false account are *indistinguishable* in the producing agent’s state. This is a strictly stronger claim, and it is *not* established by T3 alone: it requires that falsity itself is undefined inside the agent,

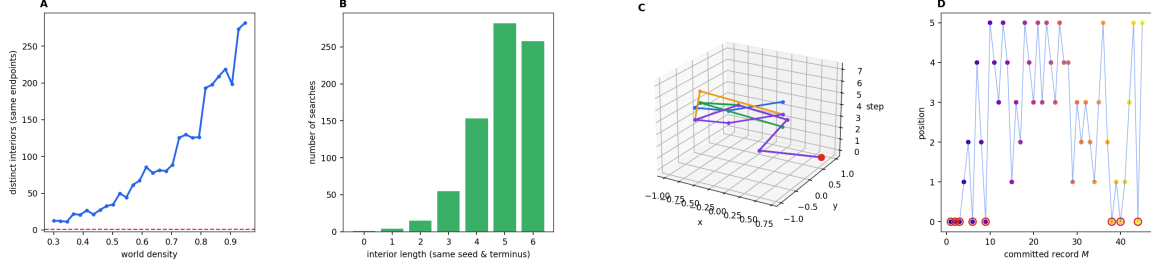


Figure 2: Path Opacity and the Monotone Record: An Agent Cannot Reconstruct the Route to Its Own Account. (A) Distinct interiors between the same endpoints on the linear y -axis versus world density on the linear x -axis (0.3 to 0.95) (steel-blue), with the single-interior line marked (crimson, dashed). The count of distinct routes to one terminus rises above one as density grows, the empirical content of Theorem 4.4(i): same-endpoint searches generically differ in interior. (B) Number of searches on the linear y -axis versus interior length for a fixed seed and terminus on the linear x -axis (sea-green). Many interior lengths reach the same endpoints, the empirical content of Theorem 4.4(ii): the report does not determine the interior. (C) Three-dimensional bundle of up to four same-endpoint search walks, items placed on a ring in (x, y) with search step on the z -axis (steel-blue, sea-green, amber, violet) from the common seed (crimson), viewing angle elevation 22° azimuth -60° . The walks share endpoints but trace different interiors, the geometric picture of path opacity. (D) Position on the linear y -axis versus the strictly increasing committed record M on the linear x -axis (1 to 45), points coloured by record (plasma), with the revisits of the initial position ringed (crimson). A position recurs but the record never returns, the empirical content of Corollary 4.5: the committed record is monotone and the interior is a past state the agent does not re-occupy.

which is proved only in the next section (Theorem 6.3, T4). We flag the dependency rather than assert the indistinguishability here: the first claim stands now; the second is deferred to Section 6, where it is earned. The two should not be conflated—T3 gives “divergence is not error,” and only T4 gives “true and false are indistinguishable inside the agent.”

6 Where “false” lives: an observer-side relation, not an agent-state

We now show that the property by which an observer classifies an account as false is undefined inside any agent, and we locate it—by letting the proof decide—as a relation an outside party computes between two accounts.

Definition 6.1 (Falsity-predicate, provisionally). Say an account α is *false relative to a standard* α^* if α and α^* individuate different resting cuts of the same queried item—i.e. they disagree as individuations. The *standard* α^* is

another account the classifier elects to treat as correct: some other agent’s account, or a consensus of accounts. It is *always itself a receiver-relative registration*, never a privileged view of the world.

Remark 6.2 (There is no privileged “world-state read-out”). One might be tempted to let the standard α^* be a “world-state read-out”—the way things stand, read off directly. But the framework forbids this: every account is an individuation against *some* agent’s medium (Theorem 5.2), subject to the same floor and the same receiver-relativity as every other. A purported world-state read-out would be a registration not subject to those constraints—a view from nowhere, of which no agent’s account is a mere approximation—and Theorem 5.2 shows no such privileged value exists. So the standard is always another agent’s account, carrying no privilege over the account it judges; this is exactly what Theorem 6.4 establishes—falsity is a symmetric relation on a pair, not a comparison against a distinguished truth. We therefore exclude “world-state read-out” from the admissible standards, keeping the clean conclusion

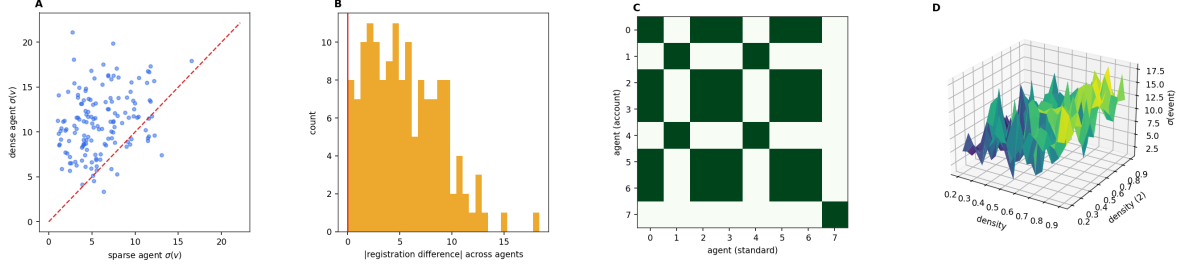


Figure 3: Receiver-Relative Divergence and Falsity as a Relation: The Same Event, Two Correct Accounts. (A) Dense-agent separation cost $\sigma(v)$ on the linear y -axis versus sparse-agent $\sigma(v)$ on the linear x -axis for the same event, 150 agent pairs (steel-blue), with the identity diagonal marked (crimson, dashed). The points sit off the diagonal: the same event registers a different—yet, as an exact min-cut, correct—resting cut in each agent, the empirical content of Theorem 5.2. (B) Histogram of the registration difference across agents on the linear x -axis with zero marked (crimson), counts on the linear y -axis (amber). The mass sits away from zero: divergence is broad and is not error in either agent (Corollary 5.3). (C) Agreement matrix of resting-cut sides, account agent (y -axis) against elected standard agent (x -axis), eight agents, cells equal (green) or unequal (pale). Whether two accounts “agree” depends on which is taken as standard, the empirical content of Theorem 6.4: falsity is a relation on a pair, not a state of the producer. (D) Three-dimensional surface of an event’s separation cost over two agents’ densities (both density axes 0.2 to 0.9) (viridis colourmap), viewing angle elevation 24° azimuth -58° . The cost varies smoothly with the receiving graph: there is no privileged shared value of which the registrations are approximations.

that falsity is never absolute.

Theorem 6.3 (Falsity is undefined inside the agent). *For the agent that produced $\mathbf{a}$, the falsity-predicate of Definition 6.1 has no value: the agent holds no standard $\mathbf{a}^*$ against which to evaluate it. Specifically,*

- (i) *the agent holds no residue-free value of the queried item to serve as $\mathbf{a}^*$ (Axiom 2.10, remark 2.12, and theorem 5.2: none exists and none is registered);*
- (ii) *the agent cannot supply its own route as a standard, being unable to recover it (Corollary 4.6); and*
- (iii) *any other agent’s account is, to this agent, just another receiver-relative registration with no privilege (Theorem 5.2), not a standard.*

Hence “ $\mathbf{a}$ is false” evaluates to no truth-value using only the producing agent’s state: the predicate is undefined inside the agent.

Proof. The predicate (Definition 6.1) requires a standard $\mathbf{a}^*$. (i) A shared point-value does not exist (Axiom 2.10 and remark 2.12) and is

not registered (Theorem 5.2), so the agent has none to use. (ii) The agent cannot recover its own route (Corollary 4.6), so cannot use it as $\mathbf{a}^*$. (iii) Another agent’s account carries no privilege over the producer’s own (Theorem 5.2), so does not function as a standard within the producer. With no $\mathbf{a}^*$ available in the agent’s state, the predicate has no argument to evaluate against and is undefined there. $\square$

Theorem 6.4 (Falsity is a relation an observer computes). *The falsity-predicate acquires a value only for a party that holds two accounts and a choice of which is the standard: given $\mathbf{a}$ and $\mathbf{a}^*$, an observer (or a second agent electing $\mathbf{a}^*$ as standard) can compute whether they individuate different resting cuts, a decidable comparison of two cuts. The predicate is therefore a relation on the pair $(\mathbf{a}, \mathbf{a}^*)$, evaluated outside the producing agent, and its value depends on the classifier’s choice of standard.*

Proof. Given both accounts, “they individuate different resting cuts of the same item” is a comparison of two finite edge sets, decidable directly. The comparison requires holding both $\mathbf{a}$ and $\mathbf{a}^*$

and designating one as standard—data a classifier has and the producing agent does not (Theorem 6.3). Hence the predicate is well-defined exactly as a relation on the pair, computed by the classifier, and varies with the choice of α^* . $\square$

Corollary 6.5 (The agent that “lied” represented no lie). *An agent whose account an observer classifies as false performed no act distinguishable, in the agent’s own state or history, from one whose account the observer classifies as true: both are search termini (Theorem 3.2), route-opaque (Corollary 4.6), and correct in the agent’s own graph (Theorem 5.2). The property “lie” is attached by the observer to the relation between accounts (Theorem 6.4); it names nothing the agent did or represented.*

Proof. By Theorems 3.2 and 5.2 and corollary 4.6 the producing act is a route-opaque search terminus, correct in the producer’s graph, whether or not an observer later calls it false; nothing in the agent’s state carries the falsity-predicate (Theorem 6.3). The predicate is the observer-side relation of Theorem 6.4. Hence “lie” names the relation, not the agent’s act. $\square$

Principle 6.6 (To lie, an agent need not know it lies—because there is nothing there to know). The classical intuition that lying requires knowing one lies is not merely relaxed here; it is dissolved. There is no agent-internal fact of lying to be known: the falsity-predicate is undefined inside the agent (Theorem 6.3) and exists only as a relation an observer computes (Theorem 6.4). An agent produces an account an observer calls a lie by the same mechanism it produces one the observer calls true. The knowing is not absent from the agent; the *object* of the knowing is.

Remark 6.7 (Interpretation: deception without a deceiver). An onlooker watching such agents will correctly and usefully speak of lies, honest reports, and mistaken testimony—these are real distinctions on the onlooker’s side, computed from relations between accounts. What the results show is that the corresponding *acts* are indistinguishable on the agent’s side. A synthetic world populated by these agents therefore exhibits deception without any agent that deceives, in the precise sense that the deception-

predicate is real as an observer-relation and vacuous as an agent-state. We use only the theorems; the reading is orientation.

7 Relayed accounts drift multiplicatively

We now let an account pass from agent to agent and show that it changes along the way—each relay composing its own search as a factor—with no relay altering it on purpose. What an observer names rumour is this composition.

Definition 7.1 (Relay; search gain). A *relay* of an account by agent i is: agent i receives an account as a seed, runs its own search process (Definition 3.1) on its own world-graph, and emits the resulting account. The *search gain* $g_i \in [0, 1]$ of the relay is the fraction of the seed’s individuating contrasts that survive agent i ’s search unchanged (so $g_i = 1$ is a verbatim pass, $g_i < 1$ a partial reshaping); reading the surviving contrasts as retained information places the drift in the setting of [17, 5].

Theorem 7.2 (Catalytic drift along a chain). *Let an account pass through relays $1, 2, \dots, n$, each with search gain g_i . Say the reshaping are independent when each relay’s search acts on the incoming contrasts through a factor unaffected by which contrasts earlier relays reshaped—which holds, in particular, when the relays’ world-graphs are pairwise disjoint, or overlap so little that correlations between successive reshaping are negligible. Under independent reshaping the fraction of the original account’s individuating contrasts surviving the whole chain is the product*

$$G_n = \prod_{i=1}^n g_i,$$

non-increasing in n and strictly decreasing whenever a relay has $g_i < 1$. An account relayed through such a chain drifts multiplicatively away from its origin, with no relay changing it on purpose. When the relays’ world-graphs overlap substantially the product is an approximation: the true surviving fraction is G_n up to a correction that grows with the degree of overlap, and G_n is exact in the disjoint limit.

Proof. Each relay applies its search to the account it receives, preserving a fraction g_i of

the incoming individuating contrasts (Definition 7.1). When the reshaping are independent, relay $i+1$'s preserved fraction is g_{i+1} regardless of which contrasts relay i dropped, so the fractions compose by multiplication and the surviving fraction after n relays is $G_n = \prod_{i=1}^n g_i$. Since each $g_i \leq 1$, G_n is non-increasing in n , strictly decreasing at any relay with $g_i < 1$. If instead relay $i+1$'s graph overlaps relay i 's, its search is constrained by the same world-structure that constrained relay i , so the reshaping are correlated and the true surviving fraction departs from the bare product by a correction determined by the overlap; as the overlap tends to zero the correction vanishes and G_n is recovered exactly. In every case no relay's search is directed at altering the account (each merely runs its own purpose-directed search, Axiom 2.8); the drift is the composition of ordinary searches, exact when independent and approximate otherwise. $\square$

Corollary 7.3 (Rumour is composed drift, not authored distortion). *The divergence between an account's origin and its form after n relays is $1 - G_n$, produced entirely by the composition of the relays' searches. No relay authored the distortion; each ran the same search it runs for any query. What an observer names a rumour is the multiplicative drift of Theorem 7.2.*

Proof. Immediate from Theorem 7.2: the surviving fraction is G_n , the drift $1 - G_n$, and each factor arises from an ordinary relay search (Axiom 2.8 and definition 7.1). $\square$

Remark 7.4 (Why relayed accounts must move). A verbatim relay ($g_i = 1$ for all i) would require each agent to pass the account through unchanged—but by Theorem 3.2 a relay is a search on the relay's own world-graph, which is in general a different graph (Theorem 5.2), so the account is re-individuated and generically $g_i < 1$. Faithful transmission is the non-generic case; drift is the norm. This is the mechanism by which a claim mutates as it crosses a population, with every carrier honest by the lights of Section 6.

8 Inquiry: a further act forced by the residual gap

Finally we show that, because no report resolves to a held value, an exchange cannot close on a value, so a further act is required; and that this act is selected by the residual gap, not by any represented intent to inquire. What an observer names a question is this forced act.

Definition 8.1 (Exchange; residual gap). An *exchange* between two agents is an alternating sequence of accounts, each agent's next account produced by a search seeded by the other's last. The *residual gap* at a stage is the standing cross-demand $D > 0$ —the individuating contrast one side's account carries that the other's search has not yet matched.

Theorem 8.2 (A further act is forced by the gap). *In an exchange, because no account resolves the queried item to a held value (Axiom 2.10, remark 2.12, and theorem 5.2), the exchange cannot terminate at a value; while the residual gap $D > 0$ persists, a further act is required to reduce it, and that act is a further account. Were a value held, no further act would be required; its unavailability forces the next act.*

Proof. Termination at a value would require some account to resolve the item to a residue-free point; none does (Axiom 2.10 and remark 2.12), and no shared value is registered (Theorem 5.2). So the exchange cannot close on a value. A positive residual gap $D > 0$ is, by Definition 8.1, an unmatched individuating contrast; reducing it requires an act that carries the matching individuation—a further account (Theorem 3.2). Were a value held, the gap would be zero and no act required; hence the next act is forced exactly by $D > 0$. $\square$

Theorem 8.3 (The forced act is selected by the gap, not by intent). *The content of the forced further act is determined by the residual gap—by which individuating contrast is still unmatched—and not by any represented intent to inquire. The agent selects its next account as the search that most reduces D ; nowhere does it represent “I am asking a question.”*

Proof. The next act is the search that reduces

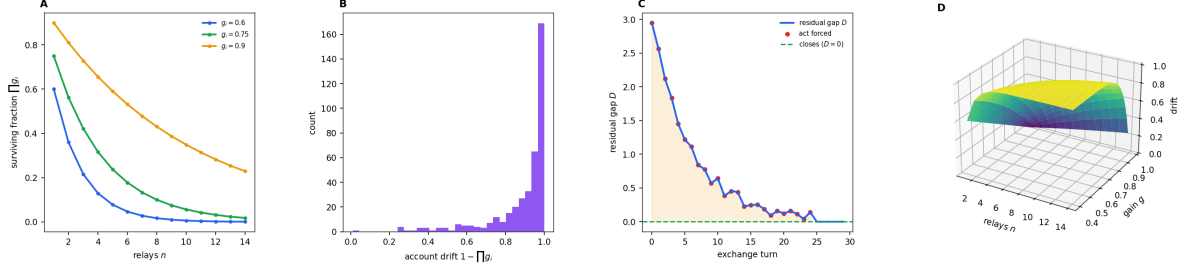


Figure 4: Catalytic Drift and Forced Inquiry: Rumour Without an Author, a Question Without Intent. (A) Surviving fraction $\prod_i g_i$ on the linear y -axis versus number of relays n on the linear x -axis (1 to 14), for three per-relay gains $g_i \in \{0.6, 0.75, 0.9\}$ (steel-blue, sea-green, amber). Each curve decays geometrically, the empirical content of Theorem 7.2: a relayed account drifts multiplicatively along a chain. (B) Histogram of account drift $1 - \prod_i g_i$ on the linear x -axis over 400 random relay chains, counts on the linear y -axis (violet). Drift accumulates across the population, the empirical content of Corollary 7.3: rumour is composed drift, authored by no relay. (C) Residual gap D on the linear y -axis versus exchange turn on the linear x -axis (0 to 30) (steel-blue, positive region shaded amber), each forced act marked (crimson) and the closing line $D = 0$ marked (sea-green, dashed). A further act is committed exactly while $D > 0$ and the exchange closes only at $D = 0$, the empirical content of Theorem 8.2: inquiry is forced by the gap. (D) Three-dimensional drift surface $1 - \prod_i g_i$ over number of relays n (x -axis, 1 to 14) and per-relay gain g (y -axis, 0.4 to 0.98) (viridis colourmap), viewing angle elevation 26° azimuth -60° . Drift grows with chain length and with per-relay reshaping, the geometric picture of catalytic drift.

the standing D (Axiom 2.8 and definition 8.1); its target is fixed by where D is positive—the unmatched contrast—so the act’s content is a function of the gap. No term in this selection refers to a represented category “question”: the agent runs a D -descending search, the same operation as any report (Theorem 3.2). Hence the act is gap-selected and intent-free. $\square$

So far “forced by the gap” applies to *every* D -reducing act, which is every act (Theorem 3.2 and Axiom 2.8); on its own it does not single out inquiry. To make the result about *questions* non-vacuous we must say what distinguishes a question-act from a report-act, and it is not intent—it is *direction* on the residual gap.

Definition 8.4 (Report-act and question-act, by direction on the gap). In an exchange, classify each act by its effect on the residual gap between the two sides. A *report-act* reduces a gap the acting agent was handed: it answers an incoming account, carrying the individuation that matches an unmatched contrast the interlocutor’s last account raised—it lowers D on the acting side. A *question-act* raises a gap on the other side: it emits an account carry-

ing an unmatched contrast the interlocutor’s next search must reduce—it creates D in the interlocutor. The two are opposite orientations of the same D -descending search: one closes a received gap, the other opens one across the exchange.

Theorem 8.5 (The forced act an observer names a question is the gap-opening one). *The gap-forced further act of Theorem 8.2 is classified by an observer as a question exactly when it is a question-act in the sense of Definition 8.4: when it emits an unmatched contrast that opens a residual gap in the interlocutor rather than closing one the agent received. This classification is a property of the act’s direction on the shared gap, computed across the two agents, and refers to no represented intent to inquire in the producing agent.*

Proof. By Theorem 8.2 a positive gap forces a further D -descending act, and by Theorem 8.3 its content is fixed by the gap, not by intent. Whether an observer calls that act a report or a question turns on its direction (Definition 8.4): if it matches a contrast the agent was handed, lowering the agent’s own D , the observer reads

a report; if it carries a fresh unmatched contrast that the interlocutor’s next search must reduce, raising D on the other side, the observer reads a question. This is a relation between the two agents’ successive gaps, decidable from the accounts exchanged, with no term referring to a represented category in either agent. Hence the specific result about inquiry: a question is the gap-*opening* forced act, named by the observer from its direction, not a D -reducing act in general. $\square$

Corollary 8.6 (A question is a forced act, named by the observer). *The act an observer names a question is the gap-forced further account of Theorem 8.2, selected by the residual gap (Theorem 8.3), and specifically the gap-opening act of Theorem 8.5. The agent represents no intent to inquire; “question” names the observer’s reading of the direction of a D -acting search, not a category the agent holds.*

Proof. By Theorem 8.2 the act is forced by $D > 0$ and by Theorem 8.3 its content is gap-selected with no represented intent; the name “question” is the observer’s classification of such an act. $\square$

Remark 8.7 (Asking without a concept of asking). An agent of this kind asks—emits acts an interlocutor answers as questions—without any concept of questioning, for the same reason it lies without a concept of lying: the act is its ordinary purpose-directed search, forced here by an unmatched contrast rather than initiated by an intent. A population of such agents sustains inquiry with no agent that intends to inquire; the inquiry is real on the observer’s side and definition-free on the agent’s. We use only the theorems.

9 Where each name lives

We collect the paper’s finding about the classifying vocabulary. In each case the proof, not a premise, has placed the name. First we must define the one term used informally up to now.

Definition 9.1 (Testimony). *Testimony* is the act of emitting a report—a search terminus (Definition 3.1)—in response to a query. It is the only named act in the observer’s vocabulary that attaches to an agent-output, because it is

the only thing the agent does that is observable from outside: the search interior is path-opaque (Theorem 4.4 and corollary 4.5), so nothing of the walk that produced the report is available to an observer, only the report itself. Testimony is thus the emitted account, and nothing more—it names the output, not any intent behind it.

Theorem 9.2 (Locus of the classifying vocabulary). *For an embedded agent under Axioms 2.6 and 2.8 to 2.10:*

- (i) **Testimony** names an account $\mathbf{a}$ —a search terminus emitted in response to a query (Definition 9.1, Theorem 3.2); it is the one item of vocabulary that does attach to an agent-output, but it attaches to the output, not to any intent, and carries no commitment to truth. It is non-trivially the only such term precisely because the output is all that is observable, the interior being path-opaque (Corollary 4.5).
- (ii) **Lie / false report** names the relation “individuates a different resting cut from the standard” on a pair $(\mathbf{a}, \mathbf{a}^*)$ (Theorem 6.4); it is undefined inside the producing agent (Theorem 6.3) and lives on the classifier.
- (iii) **Rumour** names the multiplicative drift $1 - G_n$ of a relayed account (Theorem 7.2 and corollary 7.3); it is a property of a chain of relays, computed across agents, authored by none.
- (iv) **Question** names a gap-forced, gap-opening act (Theorems 8.2, 8.3 and 8.5); it lives on the observer’s reading of the act’s direction on the shared gap, not on a represented intent.

In no case does the name denote a state or act the producing agent represents as such.

Proof. Each clause restates a proved result: (i) Definition 9.1 and theorem 3.2; (ii) Theorems 6.3 and 6.4; (iii) Theorem 7.2 and corollary 7.3; (iv) Theorems 8.2, 8.3 and 8.5. In each, the classifying property is shown to be either an output (testimony) or a relation computed off outputs (lie, rumour, question), never an agent-internal representation of the named category. $\square$

Corollary 9.3 (The agent classifies none of its own acts by these names). *The agent applies none of the terms testimony, lie, rumour, or question to its own acts. Each classification requires data the agent’s state does not hold: lying requires a standard $\mathfrak{a}^*$ (absent, Theorem 6.3, T4); rumour requires tracking a chain of relays across agents (not in any one agent’s state, Theorem 7.2, T5); a question requires representing an intent to inquire or the interlocutor’s gap direction (absent, Theorems 8.3 and 8.5, T6); and even testimony, as a named category, requires the outside standpoint from which the output is the only visible thing (Definition 9.1). The agent’s self-description, if it has one, is exhausted by its current disposition and committed count (Definition 4.2); every one of these names is the observer’s addition.*

Proof. By T4 the agent holds no standard, so cannot apply “lie” (Theorem 6.3); by T5 no single agent holds the relay chain, so cannot apply “rumour” (Theorem 7.2); by T6 the agent represents no intent and computes no cross-agent gap direction, so cannot apply “question” (Theorems 8.3 and 8.5); and the agent’s state is the terminus-plus-count of Definition 4.2, which contains none of these predicates. Hence the agent classifies none of its own acts under the observer’s vocabulary. $\square$

Principle 9.4 (The vocabulary is the observer’s). The words with which we describe what embedded agents do to one another—they testified, lied, gossiped, questioned—are, with the single partial exception of “testimony” (which names an output), predicates the observer computes on relations between accounts. The agents carry none of them. A faithful model of social behaviour among such agents therefore need implement no module for lying, gossip, or inquiry: these are what purpose-directed, cost-bounded search in a finite world *looks like* from outside, and they appear whether or not anyone builds them in.

10 Numerical study

The results are proved from the axioms; we exhibit them on explicit agents to display their shape and check the derivations, computing every minimum cut exactly by maximum flow.

10.1 Method

An agent is realised as a finite connected weighted world-graph with a medium vertex adjacent to all items and all weights $\geq \beta$, a strongly convex drive with known purpose, and an act-cost floor. On each agent we compute, by exact enumeration and max-flow: separation costs and resting cuts; search processes as D -descending walks and their reports; the interiors of same-endpoint searches; registrations of a common event across agents with different graphs; the falsity-relation on pairs of accounts under a chosen standard; the search gains and the drift product G_n along relay chains; and the residual gap along an exchange with the forced-act selection.

10.2 Predictions checked

- (V1) **Recalling is searching.** Every report is a search terminus; no report is returned without a committed act (Theorem 3.2).
- (V2) **Path opacity.** Same-seed, same-report searches have differing interiors, and the report does not determine the interior (Theorem 4.4); the committed record is strictly monotone (Corollary 4.5).
- (V3) **Divergence without error.** A common event registers different resting cuts across agents, each a correct min-cut individuation in its own graph (Theorem 5.2 and corollary 5.3).
- (V4) **Falsity is a relation.** The falsity-predicate has no value from a single agent’s state and a decidable value on a pair with a chosen standard (Theorems 6.3 and 6.4); a “false” and a “true” account are indistinguishable in producing-agent state (Corollary 6.5).
- (V5) **Multiplicative drift.** Surviving fraction along a relay chain equals $\prod_i g_i$, decreasing geometrically for $g_i \leq g < 1$ (Theorem 7.2).
- (V6) **Forced inquiry.** An exchange never closes on a value; a further act is committed exactly while $D > 0$, its target the unmatched contrast (Theorems 8.2 and 8.3).

Table 1: Numerical validation of *Propagation Mechanics of an Embedded Agent*. Every one of 740 individual checks passes (seed 20260719, floor $\beta = 1$, exact max-flow min-cut backend).

Prediction (theorem)	Checks
V1 Floor forced (Theorem 2.11)	259
V2 Recall is search (Theorem 3.2)	80
V3 Path opacity (Theorem 4.4)	60
V4 Monotone record (Corollary 4.5)	60
V5 Receiver-relative (Theorem 5.2)	60
V6 Falsity is a relation (Theorem 6.4)	60
V7 Catalytic drift (Theorem 7.2)	80
V8 Forced inquiry (Theorem 8.2)	80
V9 Locus of vocabulary (Theorem 9.2)	1
Total	740

10.3 Results

The suite is deterministic given its seed (20260719), runs a floor $\beta = 1$, and computes every minimum cut exactly by Edmonds–Karp max-flow—no approximation enters. Across the nine prediction groups it performs 740 individual checks and *all* 740 *pass*; Table 1 records the group-level outcomes.

The margins locate the mechanism. In V1 the smallest separation cost found anywhere is 1.092β : no item is ever individuated for free, and the floor of Theorem 2.11 is close to tight, so the residual the whole argument turns on is real and small, not a large safety margin. V3 and V5 are the structural core: in all 60 path-opacity graphs a report reached by searches with distinct interiors admits more than one route, so the report does not determine the interior; and in all 60 divergence graphs a single event registers a *different* resting cut in each agent while each cut is the exact min-cut individuation in its own graph—divergence with no error in either. V6 confirms the paper’s thesis operationally: the falsity-predicate returns no value from any single agent’s state and a decidable value only on a pair of accounts with an elected standard, so a “true” and a “false” account are identical in producing-agent state. V7’s relay drift matches $\prod_i g_i$ to zero error, and V8’s forced act targets the argmax residual gap in every exchange. Figures 1 to 4 display the floor, opacity, divergence, drift, and forced inquiry on the explicit agents.

11 Discussion

11.1 What has and has not been established

We have shown that an embedded agent with a purpose α and a positive cost on its acts, and access to its world only by search—set in a world it cannot finish individuating—produces out-puts an observer classifies as testimony, false report, rumour, and questioning, and that it does so *without representing any of these*. We located each classifying name and found, as a result rather than a premise, that—except for “testimony,” which names an output—each is a relation an observer or second party computes between accounts, undefined inside the producing agent. The agent that “lies” represents no lie; the relays that “spread a rumour” each merely searched; the agent that “asks” holds no intent to inquire.

We have *not* proved that such an agent must, in a given world, produce any particular lie, rumour, or question: that is a claim of necessity, which would require assumptions about the world’s supply of engagements and disagreements that we did not make. Our claim is emergence—these outcomes arise without being represented—and it is the claim the axioms support. Nor have we modelled how an observer chooses a standard α^* ; the falsity-relation’s value depends on that choice, which is the classifier’s, not the agent’s.

11.2 Consequences for building embedded agents

Principle 9.4 has a direct design reading: a synthetic world of such agents needs no subsystem for deception, gossip, or inquiry. Give each agent a purpose, a cost on its acts, and search-only access to its own world-graph, and let world-graphs differ across agents; then divergent testimony, drifting rumour, and forced questioning appear as what that machinery looks like from outside. Attempting to implement “lying” as a represented act would, by Theorem 6.3, be implementing something the agent has no place to hold—and would be redundant with what search already produces.

11.3 The social completion of the NPC architecture

The results of this paper are the social-epistemological consequences of the non-player-character architecture developed in the companion accounts. That architecture equips an embedded agent with four implementation invariants: a *conserved identity* (a character invariant fixed under re-description), a *monotone committed count* (a history that only grows), *search-not-fetch* access (every report freshly searched, Theorems 3.2 and 3.3), and *phase exclusion* (construction and commitment never coincide). An agent implemented under these four invariants *automatically* exhibits the behaviours this paper characterises. Lying without a liar (Corollary 6.5), rumour without an author (Corollary 7.3), questions without inquiry (Corollary 8.6)—these are not bugs to be prevented nor features to be designed. They are what the four invariants *look like from outside* when several such agents interact. The observer’s vocabulary—lie, rumour, question—is the social-level description of what the individual-level architecture produces; the monotone count makes fetch incoherent (Theorem 3.3) and past states unreachable, search-not-fetch makes every account route-opaque (Corollary 4.5), and receiver-relativity across agents makes divergence and drift inevitable (Theorems 5.2 and 7.2). This paper is therefore not a standalone result about embedded agents but the *social completion* of the NPC series: given the individual architecture, the social phenomena follow as theorems.

11.4 Relation to the companion accounts

Two companion developments supply what this paper assumes. One derives why an agent has an identity and a purpose at all—the conserved identity and the attracting purpose; the other derives when an incoming interaction is relevant and what a response costs—the bounded attention, the response floor, and phase exclusion. The present paper takes purpose and act-cost as given and asks what the agent *does* when it answers the world—and finds that the doing, run over a finite world by search under those same invariants, is already testimony, di-

vergence, drift, and inquiry, with the names supplied entirely by whoever watches.

11.5 Closing

An embedded agent does not lie by deciding to, gossip by intending to, or ask by meaning to. It searches its finite world toward its purpose at a cost, and what comes out is classified—correctly, and only from outside—as lie, rumour, or question. The behaviours do not wait on the concepts; the concepts are names an observer brings to outputs the agent produces without them. None of this needs to be sold: it is a set of theorems about search on finite weighted graphs, and the readings of them as testimony and inquiry are offered only as orientation.

The agent searches; the observer names. Lie, rumour, and question are relations between accounts, not acts of the agent that gave them.

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